

Identifying, Capturing, and Reusing Tacit Knowledge in Creative Domains with Generative AI

Kihoon Son

School of Computing

KAIST

Daejeon, Yuseong-gu, Republic of Korea

kihoon.son@kaist.ac.kr

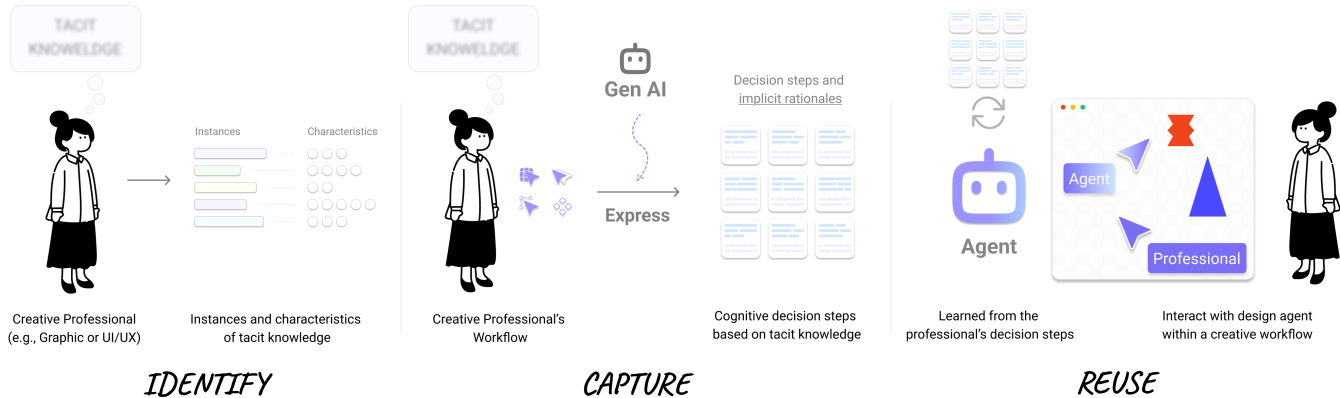


Figure 1: Overview of this dissertation. My research identifies the types of tacit knowledge held by creative professionals in the graphic or UI/UX domains, expresses this knowledge through generative AI, captures the expressed knowledge, and envisions a new type of creative collaboration with personalized design agents—trained on the captured knowledge.

Abstract

Creative professionals—such as graphic and UI/UX designers—rely heavily on tacit knowledge: intuitive, context-specific insights that are often difficult to articulate. This implicit nature makes design expertise hard to capture, share, or reuse, limiting both individual learning and the development of AI systems that meaningfully support creative work. My Ph.D. research explores how tacit knowledge can be surfaced and reused within creative workflows to enhance AI-supported design practice. I investigate when and how tacit knowledge manifests in design processes, how it can be externalized and captured, and how it can be reused to support designers in AI-enhanced environments. To this end, I combine qualitative inquiry, the development of generative AI-based interactive systems, and the design of AI agents to explore representations of tacit knowledge and their reuse. This research contributes to human-centered AI by advancing methods for capturing, expressing, and reusing tacit human knowledge in creative workflows.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Tacit Knowledge, Workflow Capture, Knowledge Representation, Generative AI, Creative Workflow, Human-AI Interaction, Human-Agent Collaboration

ACM Reference Format:

Kihoon Son. 2025. Identifying, Capturing, and Reusing Tacit Knowledge in Creative Domains with Generative AI. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3746058.3758467>

1 Introduction

Tacit knowledge [7, 8]—intuitive, contextually embedded, and shaped by heuristics—plays a crucial role in creative decision-making, helping to produce high-quality outcomes and improve creative workflows [2]. This type of knowledge is fundamental to expertise in creative fields such as graphic or UI/UX design (e.g., sensing the harmony between text and image or judging visually comfortable margin space), yet remains difficult to articulate, share, or formalize. Capturing and reusing this knowledge is essential not only for supporting novices' learning but also for developing AI systems capable of meaningfully assisting professional creative work.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '25, Busan, Republic of Korea

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758467>

However, capturing tacit knowledge in creative workflows presents inherent challenges. Because design decisions are often highly implicit, subjective, and deeply tied to specific contexts [4], it is difficult to formalize or represent them in standardized ways. As a result, this knowledge remains difficult to translate into reusable forms that can inform both novice training and AI-assisted design tools.

Recent advances in generative AI—such as large language and vision models—offer new opportunities to support the externalization of tacit knowledge. These models, trained on large-scale datasets, can generate plausible rationales or heuristics that prompt designers to reflect on and articulate the reasoning behind their creative choices. In this way, model-generated outputs can serve as reflective prompts that help surface and refine previously implicit intentions in their decisions. While such outputs may sometimes be inaccurate or overly generic, their expressiveness and context-aware generation capabilities can serve as useful scaffolds for externalizing tacit knowledge within design workflows.

Therefore, my dissertation focuses on three research questions that explore how to identify, capture, and reuse tacit knowledge:

- 1) What forms does tacit knowledge take in creative professionals' practice?
- 2) How can this knowledge and related design decisions be expressed and captured within their workflows using generative models?
- 3) How can the captured knowledge be reused to support the professionals in their workflows?

First of all, to uncover specific instances of tacit knowledge in design practices, I redefined its characteristics through a literature review, collected and annotated examples via interviews with designers, and analyzed their coverage in existing support systems. Based on these findings, I proposed interactive system design guidelines for capturing and reusing tacit knowledge and related design decisions in professional creative workflows.

Building on the proposed guidelines, I developed two interactive systems to explore how generative models can help designers express and capture tacit knowledge within their workflows. The first system, *GenQuery*, supports designers in externalizing early-stage ideation and exploration by using LLMs and Diffusion models. The second system, *CLEARFAIRY*, captures designers' cognitive decision steps and underlying rationales in a design workflow. Through these systems, my dissertation demonstrates that generative models can facilitate the articulation of implicit decisions and the surfacing of tacit knowledge embedded in workflows.

Finally, I envision two future directions for leveraging captured knowledge. First, I aim to develop an AI agent framework that learns from designers' decision-making processes to autonomously perform creative tasks. Second, I explore how such an agent could collaborate with designers as an interactive partner, sharing reasoning and supporting the co-creation process.

In summary, my dissertation investigates how to capture and reuse human knowledge embedded in creative workflows to support design practice through AI-based systems. I demonstrate how an HCI researcher can tackle ill-defined problems—such as capturing tacit knowledge—by beginning with qualitative inquiry, translating insights into interactive systems, and embedding them into AI frameworks or tools. Through this journey, I hope to illustrate

how HCI researchers—myself included—can meaningfully bridge qualitative inquiry, system design, and AI to engage with complex, real-world challenges in creative practice.

2 Identifying Tacit Knowledge in Graphic Design with Creative Professionals

To effectively capture tacit knowledge in creative domains such as design, my first research step is to identify the instances and characteristics of such knowledge within the design. In the initial phase of my Ph.D. research, I conducted qualitative research approaches—including interviews with professional designers, annotation studies, and a systematic literature review—to uncover the characteristics and instances of tacit knowledge in graphic design.

In this work [12], I collected 123 instances of tacit knowledge from 10 professional graphic designers (e.g., Predicting design error cases or Combining aesthetically pleasing colors) and analyzed how this knowledge manifests in practice. Specifically, I identified the key elements where tacit knowledge is revealed (e.g., colors, typography, layout), the actions through which it is expressed (e.g., creating, arranging, adjusting), and the purposes that underlie these actions (e.g., achieving visual harmony, aligning with design trends, considering user experience).

The analysis revealed that tacit knowledge in graphic design is primarily applied to **inner design elements** (e.g., Color, Font, Image, or Space) through **manipulation and cognitive actions** (e.g., Possess, Feel, Predict, or Judge), often **aiming to create visually pleasing outcomes while addressing audience needs**. However, many of these tacit decisions rely heavily on intuition and experiential knowledge, making them difficult to articulate, codify, or share.

To better understand this challenge, I systematically examined how well existing interactive design support tools have supported—or failed to support—these types of tacit knowledge. By identifying unsupported instances and understanding the reasons behind this gap—such as the high reliance on personal intuition or context-specific decision-making—I proposed design guidelines for capturing and applying tacit knowledge with an interactive system in graphic design workflows. These guidelines focus on capturing implicit decision-making through designers' actions and reasoning within their workflows, and on how such captured knowledge can be embedded into novices' workflows via interactive systems to facilitate effective knowledge transfer.

This foundational work of my dissertation lays the groundwork for building interactive systems that not only assist designers but also learn from and leverage their tacit knowledge to support creative decision-making in more human-like, context-sensitive ways.

3 *GenQuery*: Expressing Implicit Design Idea Exploratory Intent

With the advancement of generative models, their ability to generate visual and textual content offers new opportunities to help designers express their knowledge, intent, or ideas more effectively. Building on this, my second research project explores how large generative models (e.g., LLM or Diffusion) can support designers in articulating their implicit ideas or exploratory intents during the creative ideation process.

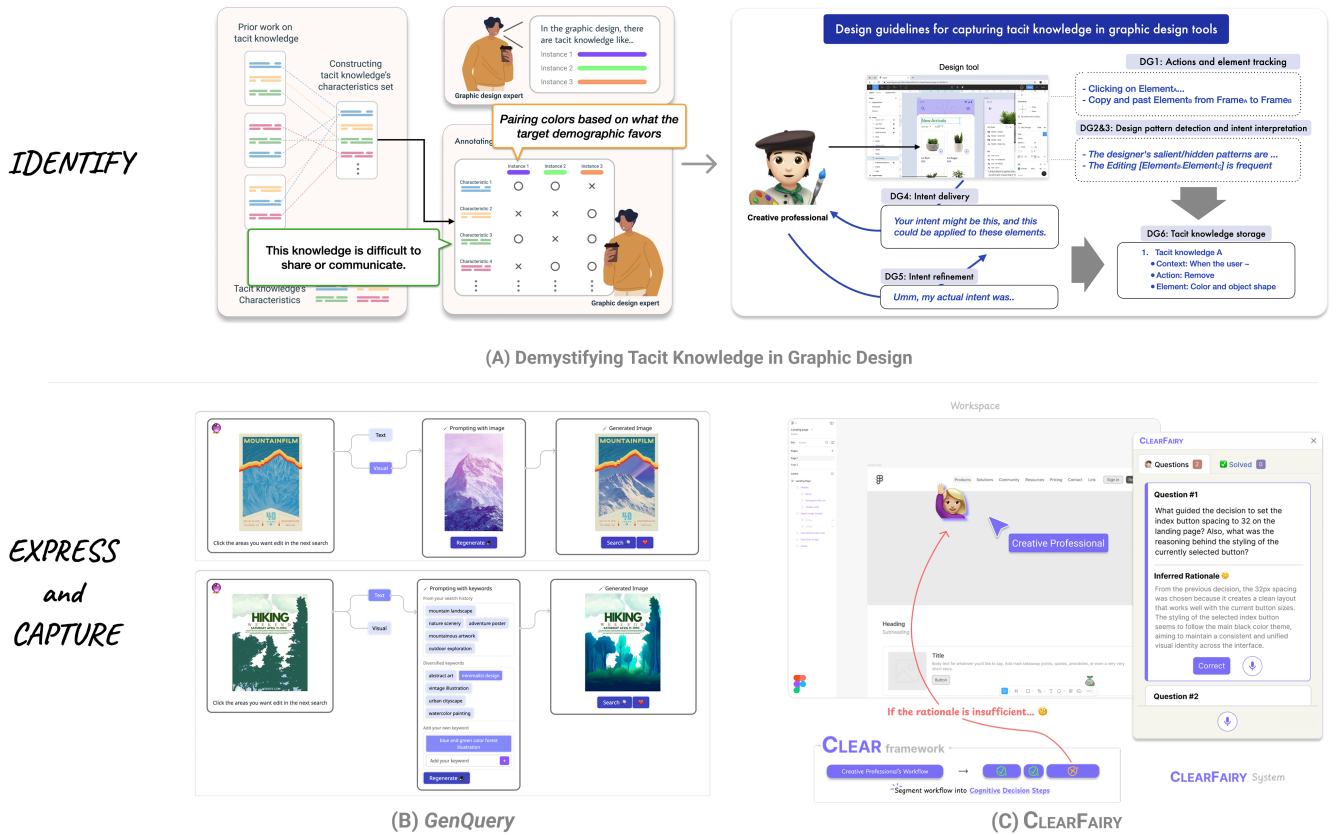


Figure 2: Three works have been conducted to identify, express, and capture tacit knowledge. This work [12] uses qualitative methods to identify instances and characteristics of tacit knowledge in graphic design. Building on these findings, it proposes design guidelines for interactive tools to help capture such knowledge during design workflows. Research (B) and (C) explore the use of generative AI to express and capture professionals’ implicit intentions and decision-making processes during design workflows. (B): *GenQuery* [13], introduces a system that supports designers in articulating their exploratory design intentions. (C): *CLEARFAIRY* [14], captures designers’ decisions and rationales by analyzing their verbal and behavioral cues within a design tool. The system also captures implicit rationales through a clarification Q&A process that addresses missing or under-specified rationales.

I developed *GenQuery* [13], an interactive system that integrates LLMs and Text-to-Image (T2I) models to assist designers in expressing and navigating their design ideas. *GenQuery* helps designers by suggesting more concrete exploratory directions for vague text queries, enabling image-based modifications to express specific visual intent, and offering keyword-based image modifications to diversify idea exploration based on prior search history.

In a user study with 16 designers, *GenQuery* allowed designers to express their design intent more clearly and discover more diverse and creative ideas compared to a baseline system without generative features. The system also enabled a new exploratory design pattern, where designers generated images as intermediate ideas to guide further exploration. While some participants noted challenges in controlling the exact outputs of the generative models, many found the unpredictability of the results helpful for sparking inspiration and avoiding creative fixation.

Through this work, I explored that generative AI can serve not only as tools for articulating implicit design ideation intent but also as triggers for creative exploration within interactive systems.

4 CLEARFAIRY: Capturing Cognitive Decision Steps in Design Workflow

To further explore how tacit knowledge can be captured in a creative workflow, my third research focuses on capturing and documenting UI/UX designers’ implicit cognitive decision steps with the underlying rationales embedded in their creative design workflow.

Traditional approaches, such as the think-aloud protocol—where users verbalize their thoughts while working—have been widely used to capture reasoning in design tasks [6]. However, such methods often impose a high cognitive load, disrupt the natural workflow, and struggle to elicit the implicit rationales behind intuitive or experience-driven decisions.

To address this challenge, I first propose the CLEAR Framework, which captures and analyzes decision-making in creative workflows by dynamically linking users' explanations, actions, and artifacts within design tools into flexible, context-aware cognitive decision steps that reveal how designers reason through creative tasks. Building on this framework, I introduce CLEARFAIRY [14], an interactive system that detects decision steps with insufficient or missing rationales and prompts designers with context-aware clarification questions. To further reduce the cognitive burden of knowledge sharing, CLEARFAIRY leverages generative models to infer possible rationales based on prior responses, allowing designers to validate, refine, or elaborate on these suggestions.

Through a user study with eight UI/UX designers, CLEARFAIRY demonstrated its effectiveness in capturing both explicit and implicit design rationales, including intuition-driven and convention-based decisions that are typically difficult to articulate. Compared to a baseline think-aloud approach, CLEARFAIRY significantly improved the quality of captured rationales and reduced the cognitive effort required for self-explanation. The captured cognitive decision steps not only provided a detailed account of the decision-making process but also offered a reusable knowledge unit that could be utilized in AI-based design support tools.

This work illustrates how generative model interventions can scaffold designers' think-aloud processes, while also helping to capture the reasoning behind decision steps by suggesting or inferring rationales. In doing so, it makes tacit knowledge more accessible, interpretable, and actionable within real-world creative workflows.

5 Future Directions

My future direction envisions building a new AI agent architecture and interaction framework that can reason, create, and collaborate in ways similar to a creative professional who shares their cognitive decision-making process.

5.1 Building Agent Framework Based on the Captured Decision Steps

First, my research will explore how an agent architecture should be designed to incorporate a creative professional's cognitive decision steps. Observing the decision steps captured in the CLEARFAIRY study reveals that after making a design decision (e.g., "emphasize this text"), professionals follow their own sequence of actions to implement that decision (e.g., selecting the text, applying bold styling, and adjusting the font size). However, such human decision-making differs significantly from current agent architectures. Existing react structure-based agents [16] typically make a decision (i.e., reasoning) based on the current task environment and execute a single action among their action space. While prior work has explored enhancing reasoning for action selection using task reflections [11] or previous work trajectories [17], there has been little exploration of agent architectures that generate a sequence of actions corresponding to a decision. Therefore, in my future work, I aim to develop an agent that learns from human decision steps to autonomously generate and execute a coherent sequence of actions in response to a given decision. In other words, I propose an agent architecture capable of defining and performing effective action sequences from primitive actions to accomplish design tasks.

5.2 Towards Collaborative Design Agents: Sharing Decisions and Reasoning in Interactive Design Tools

Second, I plan to propose an interaction framework for how such an agent should interact with creative professionals within canvas-based design tools like Figma, leveraging basic actions (e.g., creating a rectangle, moving text, changing the properties of a shape). In creative workflows, professionals gradually implement their ideas within the tool, which fosters important aspects such as reflection-in-action [10], creativity [5, 9], and a sense of agency in the design process [1]. However, current human-agent collaboration primarily relies on multimodal prompts (e.g., text, images, or design files)-based turn-by-turn interaction where the agent generates a design and the professional iteratively requests modifications. This process differs from actual design workflows where creative professionals engage directly with low-level operations to construct their work. In this second line of research, I aim to explore how professionals and agent can share their decisions and reasoning units within real-time tool use, and to propose a human-agent collaboration framework in which the agent not only produces design outcomes but also provides proactive support throughout the creative process.

6 Conclusion and Long-Term Goals

Through my Ph.D. research, I have explored methodologies for capturing and reusing creative professionals' tacit knowledge using generative models. I believe this work opens new possibilities for personalized AI support grounded in captured human knowledge. If an AI system or AI-based interactive tool can understand and learn a person's tacit knowledge or implicit reasoning patterns, this knowledge can be reused to meaningfully support individuals in their creative workflows.

From a knowledge transfer perspective, I envision a future where AI agents learn and adapt to individual tacit knowledge across tasks and domains, enabling knowledge sharing not only between humans, but also between humans and AI agents, and even among agents themselves. By interacting with agents that embody expert knowledge, novices could engage in learning experiences such as Over-the-Shoulder Learning [15] or Cognitive Apprenticeship [3] more easily and at lower cost.

Building on my Ph.D. research and as part of my future academic career, I aim to continue exploring how to better capture deeply implicit, context-embedded human knowledge and how this knowledge can be reused across different domains and tasks.

Acknowledgments

I appreciate my advisor, Juho Kim, for his support and guidance on my research. I also want to thank my mentors, collaborators, friends, and lab mates for their encouragement and support. This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (Ministry of Science and ICT) (No.RS-2025-00557726). This work was also supported by Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No.2021-0-01347, Video Interaction Technologies Using Object-Oriented Video Modeling).

References

- [1] Albert Bandura. 1997. *Self-efficacy: The exercise of control*. Vol. 11. Freeman.
- [2] Marcelo Bernal, John R Haymaker, and Charles Eastman. 2015. On the role of computational support for designers in action. *Design Studies* 41 (2015), 163–182. doi:10.1016/j.destud.2015.08.001
- [3] Allan Collins, John Seely Brown, Ann Holum, et al. 1991. Cognitive apprenticeship: Making thinking visible. *American educator* 15, 3 (1991), 6–11.
- [4] Nigel Cross. 1982. Designerly ways of knowing. *Design Studies* 3, 4 (1982), 221–227. doi:10.1016/0142-694X(82)90040-0 Special Issue Design Education.
- [5] Gabriela Goldschmidt. 1991. The dialectics of sketching. *Creativity research journal* 4, 2 (1991), 123–143.
- [6] Rebecca Krosnick, Fraser Anderson, Justin Matejka, Steve Oney, Walter S. Lasecki, Tovi Grossman, and George Fitzmaurice. 2021. Think-Aloud Computing: Supporting Rich and Low-Effort Knowledge Capture. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '21*). Association for Computing Machinery, New York, NY, USA, Article 199, 13 pages. doi:10.1145/3411764.3445066
- [7] Michael Polanyi. 2012. *Personal knowledge*. Routledge. doi:10.4324/9780203442159
- [8] Michel Polanyi and Amartya Sen. 2009. The tacit dimension. With a new foreword by Amartya Sen. doi:10.1007/s11016-010-9328-0
- [9] Mitchel Resnick, Brad Myers, Kumiyo Nakakoji, Ben Shneiderman, Randy Pausch, Ted Selker, and Mike Eisenberg. 2005. Design principles for tools to support creative thinking. (2005).
- [10] Donald A Schön. 2017. *The reflective practitioner: How professionals think in action*. Routledge.
- [11] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems* 36 (2023), 8634–8652.
- [12] Kihoon Son, DaEun Choi, Tae Soo Kim, and Juho Kim. 2024. Demystifying Tacit Knowledge in Graphic Design: Characteristics, Instances, Approaches, and Guidelines. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 221, 18 pages. doi:10.1145/3613904.3642886
- [13] Kihoon Son, DaEun Choi, Tae Soo Kim, Young-Ho Kim, and Juho Kim. 2024. GenQuery: Supporting Expressive Visual Search with Generative Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 180, 19 pages. doi:10.1145/3613904.3642847
- [14] Kihoon Son, DaEun Choi, Tae Soo Kim, Young-Ho Kim, Sangdoo Yun, and Juho Kim. 2025. CLEARFairy: Capturing Creative Workflows through Decision Structuring, In-Situ Questioning, and Rationale Inference.
- [15] Michael B Twidale. 2005. Over the shoulder learning: supporting brief informal learning. *Computer supported cooperative work (CSCW)* 14, 6 (Nov. 2005), 505–547. doi:10.1007/s10606-005-9007-7
- [16] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.
- [17] Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2024. Expel: Llm agents are experiential learners. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 19632–19642.